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Oefeningenreeks 1E nr. 13

$A(z) = z^6 - 1$

$B(z) = (z-1)(z+1)(z-2)$

$\Rightarrow A(z) = (z-1)(z+1)(z-2)Q(z) + az^2 + bz + c$

$$\Rightarrow \begin{cases} a+b+c=0 \\ a-b+c=0 \\ 4a+2b+c=63 \end{cases}$$
$$\Rightarrow \begin{cases} a=-b-c \\ -b-c-b+c=0 \\ -4b-4c+2b+c=63 \end{cases}$$
$$\Rightarrow \begin{cases} a=-b-c \\ -2b=0 \\ -2b-3c=63 \end{cases}$$
$$\Rightarrow \begin{cases} a=-c \\ b=0 \\ c=-\dfrac{63}{3} \end{cases}$$
$$\Rightarrow \begin{cases} a=21 \\ b=0 \\ c=-21 \end{cases}$$

rest: $R(z) = 21z^2 - 21$

References